Logistic Regression

The classification algorithm logistic regression is used to assign observations to a discrete set of classes. Unlike linear regression, which produces a continuous number of values, logistic regression produces a probability value that can be mapped to two or more discrete classes using the logistic sigmoid function.

A regression model with a categorical target variable is known as logistic regression. To model binary dependent variables, it employs a logistic function.

The target variable in logistic regression has two possible values, such as yes/no. Consider how the target variable y would be represented in the value of "yes" is 1 and "no" is 0. The log-odds of y being 1 is a linear combination of one or more predictor variables, according to the logistic model. So, let's say we have two predictors or independent variables, and and p is the probability of y equaling 1. Then, using the logistic model as a guide:

$$\ln \frac{p}{1-p} = a + bx_1 + cx_2$$

We can recover the odds by exponentiating the equation:

$$\frac{p}{1-p} = e^{(a+b_{x1}+c_{x2})}$$

$$p = \frac{p = e^{(a+b_{x1}+c_{x2})}}{1+e^{(a+b_{x1}+c_{x2})}}$$

$$p = \frac{1}{1+e^{-(a+b_{x1}+c_{x2})}}$$

As a result, the probability of y is 1. If p is closer to 0, y equals 0, and if p is closer to 1, y equals 1. As a result, the logistic regression equation is:

$$y = \frac{1}{1+e^{-(a+b_{x1}+c_{x2})}}$$

This equation can be generalized to n number of parameters and independent variables as follows:

$$y = \frac{1}{-(\beta_0+\beta_1 x_1+\ldots+\beta_n x_n)}$$

Comparison between linear and logistic regression

Different things can be predicted using linear regression and logistic regression as shown in *Figure*

Linear The predictions of linear regression are continuous (numbers in a range). It may be able to assist us in predicting the student's test score on a scale of 0 to 100.

Logistic The predictions of logistic regression are discrete (only specific values or categories are allowed). It might be able to tell us whether the student passed or failed. The probability scores that underpin the model's classifications can also be viewed.

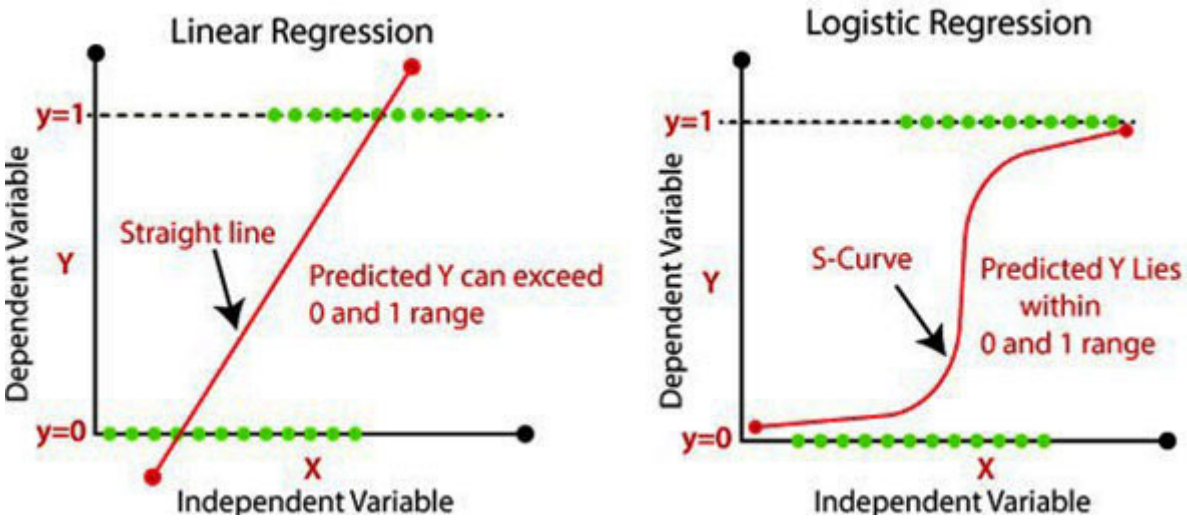


Figure 2.16: *Linear regression vs Logistic regression*

While linear regression has an infinite number of possible outcomes, logistic regression has a set of predetermined outcomes.

When the response variable is continuous, linear regression is used, but when the response variable is categorical, logistic regression is used.

A continuous output, such as a stock market score, is an example of logistic regression and predicting a defaulter in a bank using transaction details from the past is an example of linear regression.

The following *Table 2.4* shows the difference between Linear and Logistic Regression:

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Table 2.4: *Difference between Linear and Logistic Regression*

Types of Logistic Regression

Logistic regression is mainly categorized into three types as shown in *Figure*

Binary Logistic Regression

Multinomial Logistic Regression

Ordinal Logistic Regression

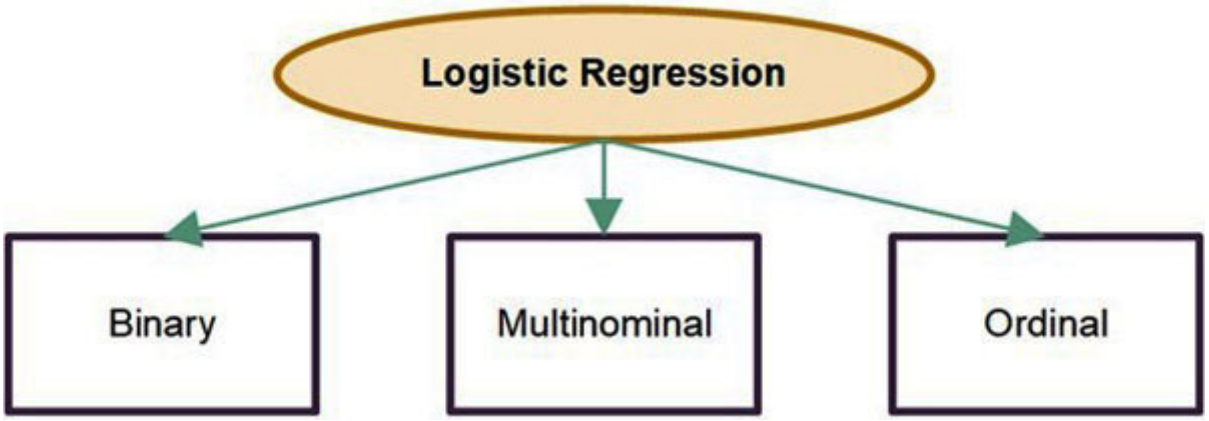


Figure 2.17: *Types of Logistic Regression*

Binary Logistic Regression

The target variable or dependent variable in binary logistic regression is binary in nature, meaning it has only two possible values.

There are only two possible outcomes for the categorical response.

For example, determining whether or not a message is a spam.

Multinomial Logistic Regression

In a multinomial logistic regression, the target variable can have three or more values, but there is no fixed order of preference for these values.

For instance, the most popular type of food (Indian, Italian, Chinese, and so on.)

Predicting which food is preferred more is an example (Veg, Non-Veg, Vegan).

Ordinal Logistic Regression

The target variable in ordinal logistic regression has three or more possible values, each of which has a preference or order.

For instance, restaurant star ratings range from 1 to 5, and movie ratings range from 1 to 5.

Examples

The following are some of the scenarios in which logistic regression can be used.

Weather Prediction regression is used to make weather predictions. We use the information from previous weather reports to forecast the outcome for a specific day. However, logistic regression can only predict categorical data, such as whether it will rain or not.

Determining Illness: We can use logistic regression with the help of the patient's medical history to predict whether the illness is positive or negative.